



try with an outlay of ₹76,000 Cr confirms the PM's vision of *AtmaNirbhar Bharat*, and positioning India as a global hub of electronics system design and manufacturing. Displays constitute a large portion of the bill of materials of many high-volume electronics products. India's display market is estimated to be \$7 billion, and it could grow up to \$19 billion by 2025. Current demands are met exclusively through imports," says Sanjay Agarwal, President of ELCINA and MD of Globe Capacitors.

Semiconductors and display units are generally referred to as the soul of modern electronics. These are also said to be among the top factors that are driving the ongoing Industry 4.0 revolution. The challenge is that both semiconductor and display manufacturing require huge capital investment, and these need to be done with the help of most modern technology. Rapid changes in technology further require investments for sustenance and gestation.

"The benefits of establishing this industry includes greater than \$11 billion dollars domestic value creation, display export opportunities of up to \$10 billion, and domestic employment generation of more than 200 thousand," adds Sanjay.

Amrit Manwani, MD, Sahasra Electronics notes, "The leadership of India, in the past six to seven years, has rightly highlighted and worked on the importance of electronics manufacturing. Several incentive schemes like EMC, SPECS, and PLI have also been introduced. But for the first time the Ministry of Electronics under the leadership of Ashwini Vaishnav and Rajeev Chandrasekhar has truly engaged with the industry to recognise the challenges and mitigate them through various reforms."

"So far, we have virtually had the absence of semiconductor manufacturing in India, even though we had strength in semiconductor design where we do designs for the

entire world. Now, because of strategic reasons and also because of the fact that our import bill will only keep on mounting if we don't have a semiconductor foundry in place or an IDM in place, this is why there is a renewed effort in bringing semiconductor manufacturing into India. There have been efforts before but what is different this time is, firstly the size of the Indian market is today standing at 3.5% of global electronics manufacturing. We expect it to become 10% of global electronics manufacturing in the next four to five years. As more and more products get manufactured in India, one way is to keep on importing the components. The other way is that, now that there are strong brands here who control the supply chains, if a fab is there, then they may well be inclined to get their memory and logic pieces out of India as well as power devices," says Saurab Gaur, Joint Secretary, MeitY.

### The government and the heavyweights

The road to making India a hub of semiconductor and display manufacturing is not going to be an easy one. Not only will the players deciding to join the game will have to take lessons from the past, but players from the equipment, materials, designing, services, and all other verticals of these two ecosystems will have to work together to make India the hub of manufacturing these.

"The global conglomerate has started seeing India as a place now. Government has walked the talk and now is the turn of the private players to showcase what they can do. Indians are very intelligent, and you may see very cost-effective TVs, smartphones, and more consumer goods getting manufactured and exported from here. It is all about backward integration and we all know what has happened in the automotive sector in the country. Hyundai exports cars at prices it

cannot achieve in its home country Korea," says Pranav.

He adds, "The scenario at the moment is now or never as the country is already on its way to have bigger exports bills of components than oil. It's not a short-term game. The horizon should be for at least 20 to 30 years."

Taiwan and China are the perfect examples of the semiconductor and display fabs being a long-term game. While the former is now a world leader in terms of semiconductor manufacturing, it still keeps investing in developing the latest tech. The latter, which started its semiconductor journey about a decade ago, is still investing in huge numbers to make sure it becomes a world leader in the next few decades. The government of China helping semiconductor companies in the country is not a hidden fact at all.

"Money going into these initiatives is hearty as the returns will take much longer time. But then it is a very stable business over the long term. You can also grow very rapidly. I have myself worked for companies which were a manufacturing start-up and they became a multi-billion dollar business over time," says Raja.

The Tata's are said to be putting more than 3.5 billion rupees for setting up an ATMP/OSAT in the country. Raja Manickam, who had earlier founded Tessolve, was roped in by Tata last year to head the project. Several other biggies from the semiconductor ecosystem have also joined Tata's in the new venture. "In my mind, all of this has to play together to make India the favourite destination for manufacturing electronics," feels Raja Manickam, CEO, Tata OSAT.

Vedanta Group joining the initiative to make displays in India is probably one of the best developments that the Indian electronics industry has come across in the recent days. Of course, the Tata's have already forayed into semiconductor packaging